Rare variant association analysis

DCEG Statistical Genetics Workshop

NIH NATIONAL CANCER INSTITUTE

November 4, 2026

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Agenda

What are rare variants	Definitions, and technologies (exome and whole genome sequencing
Importance	Examples from studies, missing heritability, drug targets, families
Associations tests	Tests: Burden, SKAT, SKAT-O, ACAT, dense and sparse associations
Non-coding rare variants	Regulatory elements, sliding windows
Incorporating annotations	Functional weights and STAAR

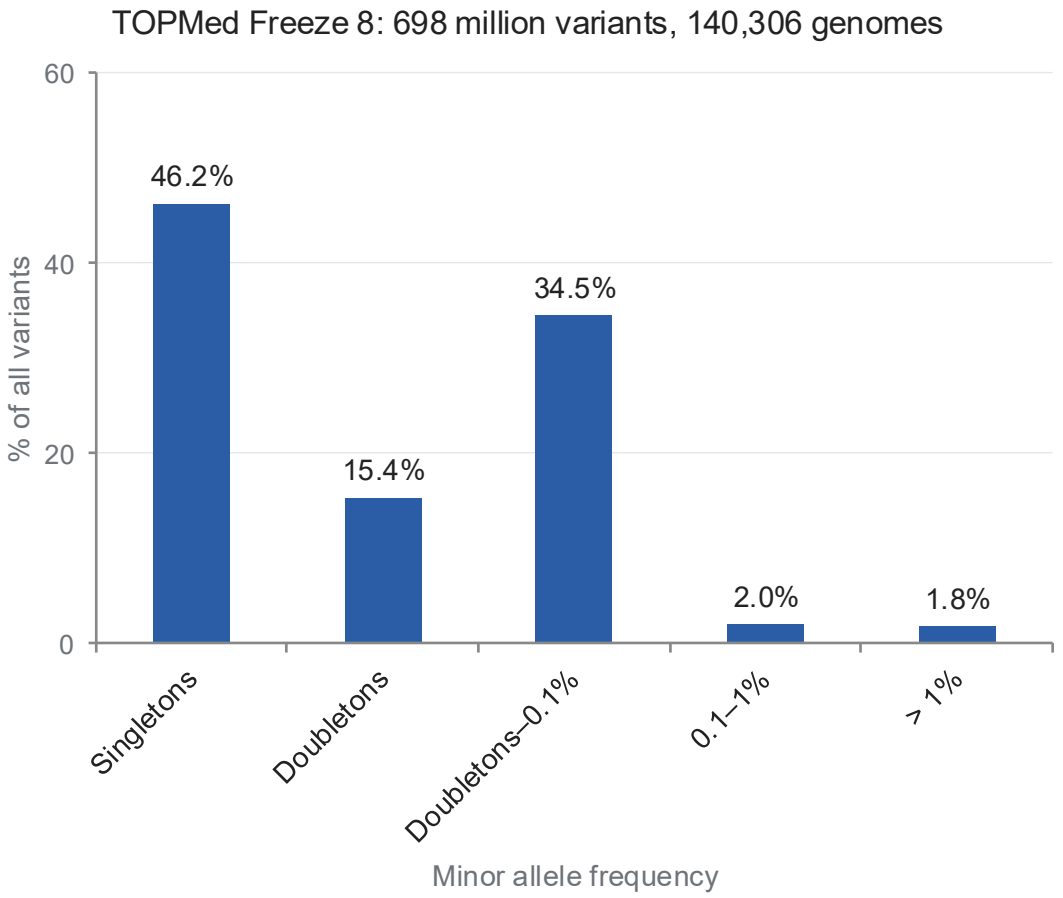
What are rare variants?

Defined by **minor allele frequency (MAF)** in the population:

- Common > 5%
- Low-frequency 1–5%
- **Rare < 1%**
- **Ultra-rare < 0.1%**: singletons and doubletons, often seen in one family or one person

Most of the variation in a genome is rare: in TOPMed, **98% of variants have MAF < 1%** and nearly half are singletons

Too few carriers to test one variant at a time, so the analysis is organised around **sets of variants**



Taliun et al. (2021) Nature, TOPMed Freeze 8

How do we find them: Sequencing

GWAS arrays + imputation: reliable down to MAF ~0.1% depending on sample size; private and population-specific alleles can be missed

Whole-exome sequencing (WES), 2010s: the ~1.5% of the genome that codes for protein; cheap, interpretable, where the first rare-variant discoveries were made

Whole-genome sequencing (WGS), now: the remaining 98%, including regulatory regions; TOPMed, All of Us, UK Biobank (490k genomes, 1.5 billion variants)

Variants per dataset in UK Biobank (n ~ 150k)



WGS sees ~5x the variants of exome + imputation combined, **almost all of it non-coding**

Image from: Gaynor et al. (2023)

Sequencing changes the **analysis problem**: millions of variants, almost all rare

Types of rare variants and why they matter

Class	What changes	Characteristics	Why it matters
Synonymous	Codon changes, same amino acid	Mostly neutral; common among coding rare variants; rarely alters splicing or translation	Built-in **negative control** and null set for burden tests
Missense (non-synonymous)	One amino acid is swapped	Anywhere from benign to damaging; can reduce or gain function; scored with REVEL, CADD, AlphaMissense	Most numerous candidate class, but **hardest to interpret**
Protein-truncating (stop-gain, frameshift, splice-site) Usually loss of function (LoF)	Transcript is cut short or exons are lost	Usually loss of function (NMD); strongest average effect; strongly depleted by selection	Natural **gene knockouts**: direction of effect and drug-target evidence (PCSK9, APOC3)
Non-coding regulatory	Promoters, enhancers, UTRs, ncRNAs	Change expression, not the protein; ~98% of the genome; defined by functional annotation	Untested territory; **needs whole genomes** and annotation-aware tests

Sequencing identifies rare variants

Class	Genotyping array + imputation	Exome sequencing (WES)	Whole-genome sequencing (WGS)
Synonymous	Poorly / not at all	Well captured	Well captured
Missense	Partial	Well captured	Well captured
Protein-truncating	Poorly / not at all	Well captured	Well captured
Non-coding regulatory	Poorly / not at all	Poorly / not at all	Well captured

Arrays

Fixed content, built for common SNPs.

Imputation is unreliable below MAF ~0.1%, so rare variants are **mostly missed**

Exome sequencing

Deep, low-cost read of the ~1–2% coding portion.

The primary data source for **rare coding** variants in earlier (versions of) biobanks

Whole-genome sequencing

Adds **non-coding** variants, structural variants and repeats.

This is becoming the standard in TOPMed and newer biobank releases

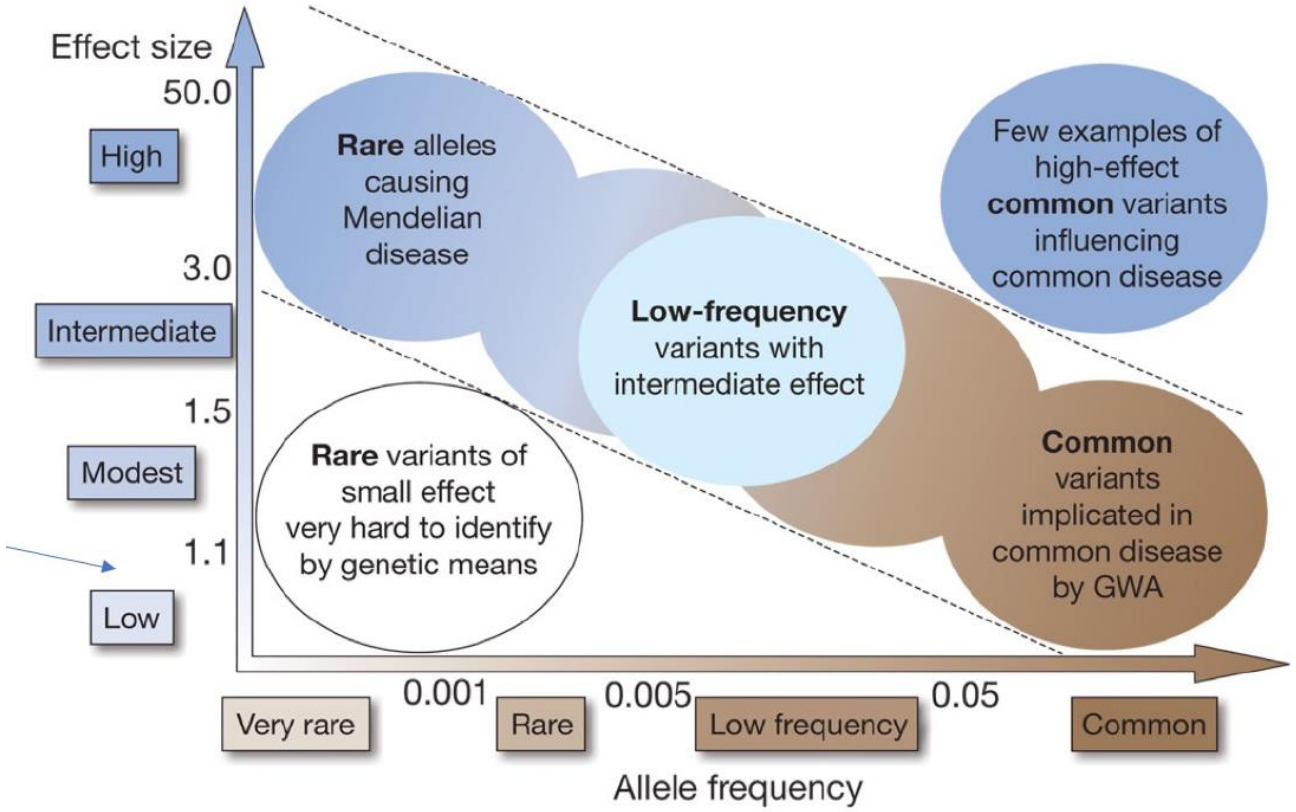
Rare-variant association studies are primarily enabled by advent of sequencing

Examples of what sequencing studies found

Gene	Rare variants	Trait	What was learned
PCSK9	Loss-of-function (~2% of African Americans)	LDL cholesterol, CHD	28% lower LDL, 88% lower CHD risk in carriers
APOC3	Four LoF alleles, combined	Triglycerides, CHD	39% lower TG, 40% lower CHD risk
HSD17B13	Splice variant	Chronic liver disease	Protective; now a drug target
BRCA1 / BRCA2	Protein-truncating	Breast, ovarian cancer	Large effects found in families before biobanks
Genome-wide	454,787 UK Biobank exomes	~4,000 traits	Thousands of gene–trait associations; rarer alleles, larger effects

A are allele that “breaks” a gene tells us which gene and which direction, something a GWAS peak rarely does

Missing heritability



GWAS loci explain only part of the heritability estimated from twins and families

Arrays can only see the **common, small-effect** corner of this map

Rare variants of **intermediate and large effect** were the proposed home of the rest

With whole genomes this can now be tested: for height, WGS-based heritability is **0.68**, compared with **~0.3** from array SNPs, with the gain coming from rare, low-LD variants

Rare variation accounts for much of what GWAS could not explain, but not all

Manolio et al. (2009) Nature

Drug targets and family studies

Natural experiments for drug targets

A loss-of-function carrier is a **lifelong human knockout** of one gene

Shows the effect of inhibiting the target, and whether that is safe, before any trial

PCSK9 LoF → PCSK9 inhibitors; ANGPTL3, APOC3, HSD17B13 followed the same route

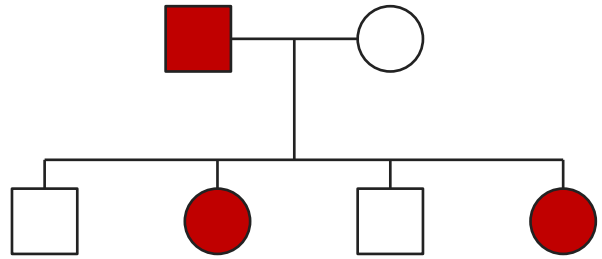
Targets with human genetic support are **~2.6x more likely** to reach approval

Family studies

A rare allele is **enriched within a family**: a variant seen once in a biobank may be carried by half of a pedigree

Large effects segregate with disease, so linkage and co-segregation can work with a few dozen people

Still one of the main routes for **Mendelian and familial cancer** genes



Filled = affected; the same rare allele in each affected relative

Main philosophy of rare variant association tests

Single variant test power depends on MAF, effect size, sample size and LD

For rare variants, bigger SE's due to low MAF at current sample sizes. Almost no detection power.

Aggregate more than one variants (in a gene or functional unit) to improve power

The simplest test: Burden

- Pick a **set** of rare variants (termed "**mask**")
  - usually exonic variants in a gene beyond a MAF cut-off
- Collapse them: count how many rare alleles each person carries
  - Also can upweight based on MAF
- Regress the trait on that single burden, as in a GWAS of one variant
- Works well when most variants in the set are causal and **act in the same direction**
- Loses power when effects are **mixed in direction** or most variants are neutral: the count then averages the signal away

Rare alleles in one gene

Sample	V1	V2	V3	V4	Burden
1	0	0	**1**	0	**1**
2	**1**	0	0	0	**1**
3	0	0	0	0	**0**
4	0	**1**	0	**1**	**2**
5	0	0	0	0	**0**
6	**1**	0	0	0	**1**

Trait ~ Burden + covariates

One regression coefficient per gene. A significantly non-zero slope is evidence that the gene matters

Li & Leal (2008) AJHG; Madsen & Browning (2009)

Input and output

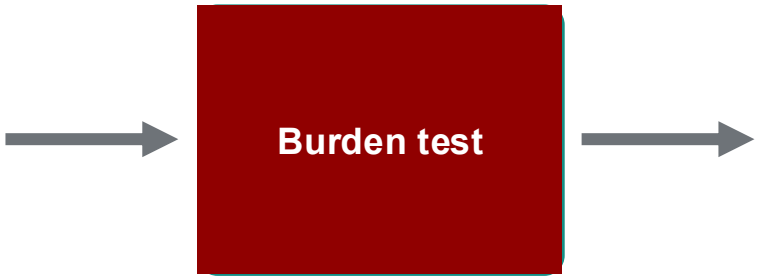
Inputs

Variant set
which gene or region, which variants (LoF, damaging missense, MAF cut-off)

Genotype matrix
n people x m rare variants; usually 0/1 (very rarely 2)

Weights
optional; to up-weight rarer alleles

Trait and covariates
continuous or binary; age, sex, ancestry PCs, relatedness



Outputs

Effect estimate
slope and SE; direction of effect

p-value
one per gene, so the exome-wide threshold is ~2.5 x 10^{-6} for 20,000 genes

Cumulative MAC
how many carriers the result rests on; below ~10 the test is unreliable

The definition of the variant set is a modelling choice and drives the result as much as the test does

Interpreting the results

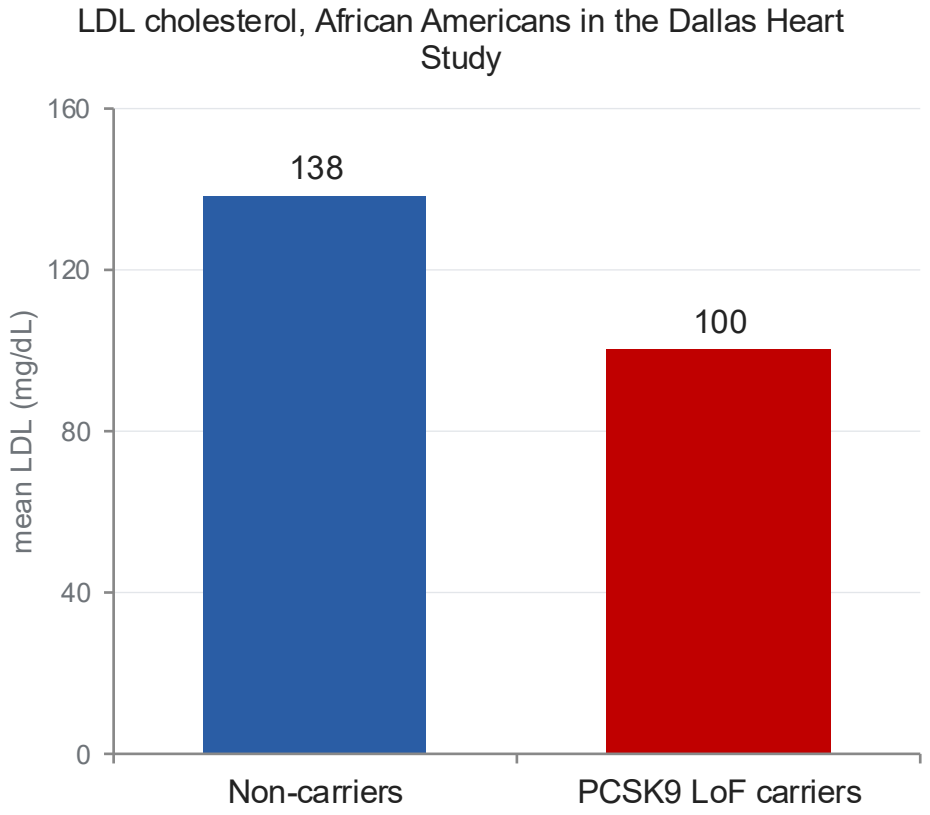
With rare alleles almost everyone has a score of 0 and a few people have 1, so the test is essentially **carriers versus non-carriers**

The coefficient is the **average shift in the trait per rare allele** in the set; its sign is the direction of effect

PCSK9 nonsense alleles: carriers' LDL is about **38 mg/dL lower** (28%), and that is what the slope reports

It is an average: a few strong alleles mixed with many neutral ones give a **diluted** slope, and opposite-direction alleles cancel

A significant burden says the **gene** matters, not which variant inside it does



Two nonsense alleles (Y142X, C679X) collapsed. 2.6% carriers

Examples of discoveries through Burden test

Gene / study	Variants collapsed	Finding	Why single-variant tests could not
APOC3, Exome Sequencing Project (2014)	4 loss-of-function alleles, 3,734 exomes	39% lower triglycerides, 40% lower CHD risk in ~1 in 150 carriers	Each allele carried by a handful of people; only the aggregate reached significance
SLC30A8, type 2 diabetes (2014)	12 protein-truncating variants, ~150,000 people	65% lower T2D risk in carriers; protective loss of function	One allele was frequent enough to test on its own and only in Finland
LDLR and APOA5, early-onset MI (2015)	Rare damaging alleles per gene, ~9,800 exomes	Carriers of LDLR alleles: 4x MI risk; APOA5: 2x	Dozens of distinct mutations, most seen once
UK Biobank exomes (2021)	Gene-level burden of pLoF and damaging missense, 454,787 people	564 genes associated with ~4,000 traits; 69 signals from singletons alone	A singleton can never be significant by itself; it can be in a burden

Aggregation turns many rare variants into one testable, interpretable signal

SKAT: Incorporate heterogeneity

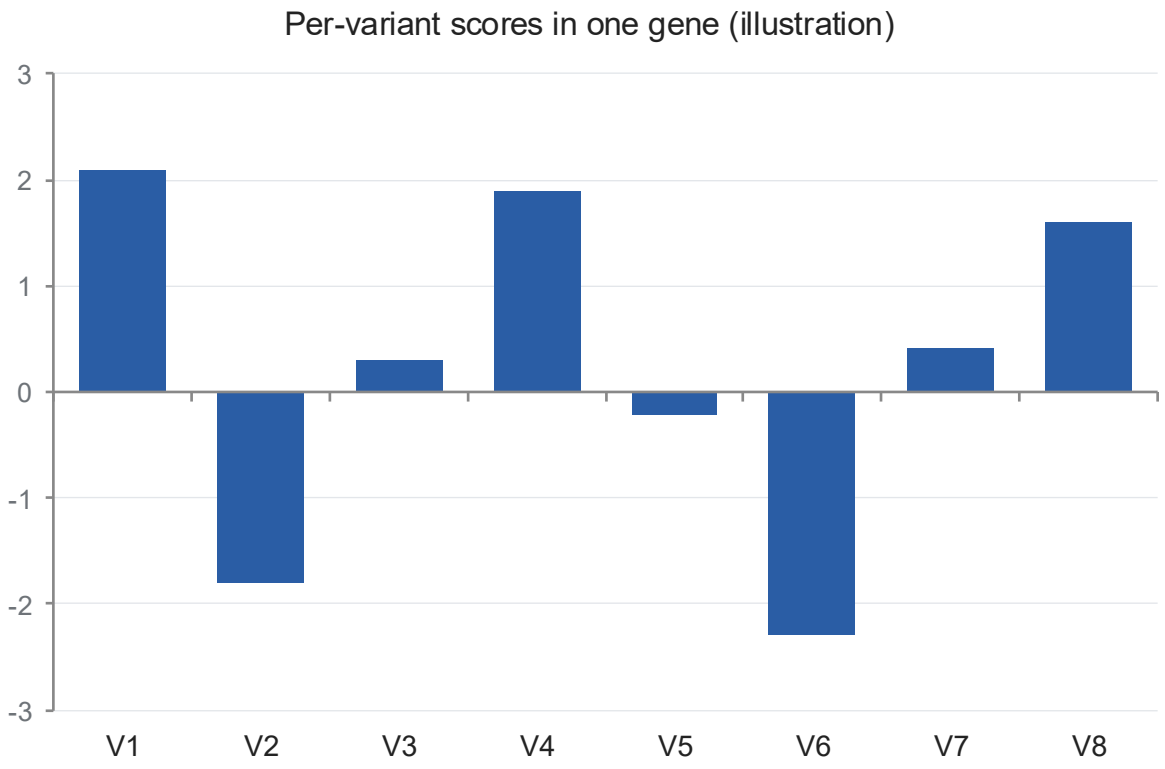
Burden: is the **average** effect of the set non-zero?

SKAT: : is the **average** effect of the set non-zero accounting for potential **heterogeneity among the effects**?

Each variant gets its own per-variant score. SKAT adds up their **squares** (weighted), so signs no longer cancel

No direction assumption, and neutral variants add little because their scores are near zero

p-value comes from a mixture of chi-squares (using Davies' method).



Burden: sum of scores	2.0	signs cancel
SKAT: sum of squared scores	**19.4**	every effect counts

Burden and SKAT side by side

Burden

Collapses first, then tests: a **linear** sum of alleles

Assumes most variants are causal and push the **same way**

Gives an **effect size** with a sign: easy to interpret and to follow up

Weakened by neutral variants and by opposite-direction alleles

SKAT

Tests per-variant effects jointly: a **quadratic** sum of scores

Allows effects in **either direction** and of any size

Gives a p-value but **no single effect estimate**

Robust to neutral variants; pays a price when burden's assumptions actually hold

	Burden	SKAT
Statistic	$(\Sigma\, w_j\, S_j)^2$	$\Sigma\, w_j\, S_j^2$
	Squared sum of scores	Sum of squared scores
Null distribution	χ^2 with 1 d.f.	mixture of χ^2

What SKAT adds over burden test

Direction: protective and risk alleles in one gene both add to the statistic

Dilution: a few real effects among many neutral variants still show up, since neutral scores are near zero

Flexibility: per-variant weights can carry MAF or functional annotation; kernels can add interactions

Speed: a score test under the null, so the model is fitted once per trait, not once per gene

Not fixed: SKAT still needs a sensible variant set, enough carriers, and large samples; with small n its p-values need a small-sample correction

Situation	Burden	SKAT
mostly causal, same direction	**best**	good
mixed directions	**fails**	**good**
few causal, many neutral	**weak**	**better**

Why incorporating heterogeneity might matter

Situation	Why SKAT does well	Example
Protective and risk alleles in one gene	Squared scores never cancel	ANGPTL4, lipid genes with gain- and loss-of-function alleles
Large sets with few causal variants	Neutral variants contribute ~0. Sometimes heterogeneity is detected.	Whole-gene masks that include all missense; non-coding windows
Sets defined without strong prior knowledge	No need to guess direction or pick only 'damaging' alleles	Regulatory regions, sliding windows across the genome
Effect sizes that differ a lot between variants	Each variant weighted by its own evidence	Mixed LoF + missense sets

Burden is preferable for curated **loss-of-function masks**, where almost every allele knocks the gene out in the same way

Sample size matters more than the choice of test: both need **thousands to tens of thousands** of people for a typical gene

What to do in practice?

ANGPTL4 and triglycerides, Dallas Heart Study

ANGPTL4 resequenced in 3,551 people; **17 rare and low-frequency variants**

E40K **lowers** triglycerides; several other alleles have weaker effects

Collapsing them adds a negative and positive effect together and the burden signal disappears

Test on the 17 variants	p-value
Burden	0.70
SKAT	**0.012**

In exome-wide scans

Biobank pipelines (SAIGE-GENE+, Genebass, REGENIE) **run burden and SKAT** on every gene and several variant masks

Burden carries most hits, because damaging alleles in one gene usually act alike

SKAT adds genes with **heterogeneous effects**: gain- and loss-of-function alleles, or a mix of damaging and benign missense

Suggestion: Run both tests. Agreement is reassuring and disagreement is sometimes informative

SKAT-O: data adaptive compromise between burden and SKAT

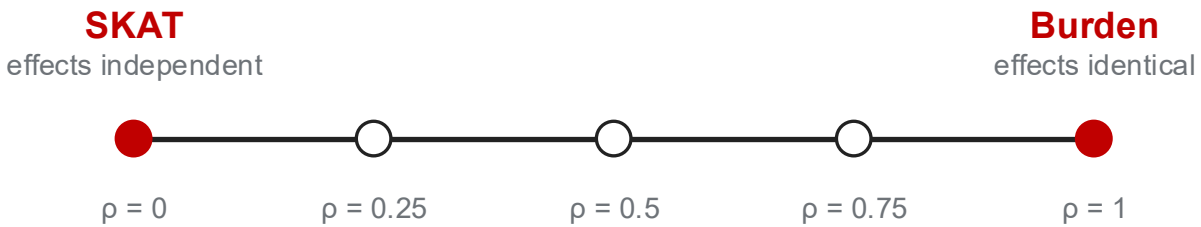
Burden and SKAT are two ends of one family: a weighted mix of the two statistics, with weight **ρ** on burden and **1 − ρ** on SKAT

ρ = 1 assumes all variants share one effect; ρ = 0 assumes effects are unrelated; values in between allow partial correlation

SKAT-O computes the test over a grid of ρ, takes the **smallest p-value**, and corrects for having searched

A slightly lower power compared with the “best” test chosen in advance but is never badly wrong

This is why SKAT-O is the **default** gene-based test in most sequencing pipelines



SKAT-O scans this line and reports the best-supported point

ρ is the assumed correlation between the effects of the variants in the set

One p-value per gene that is close to the best of both tests whatever the architecture

Lee et al. (2012) Biostatistics

Thoughts on designing a rare-variant study

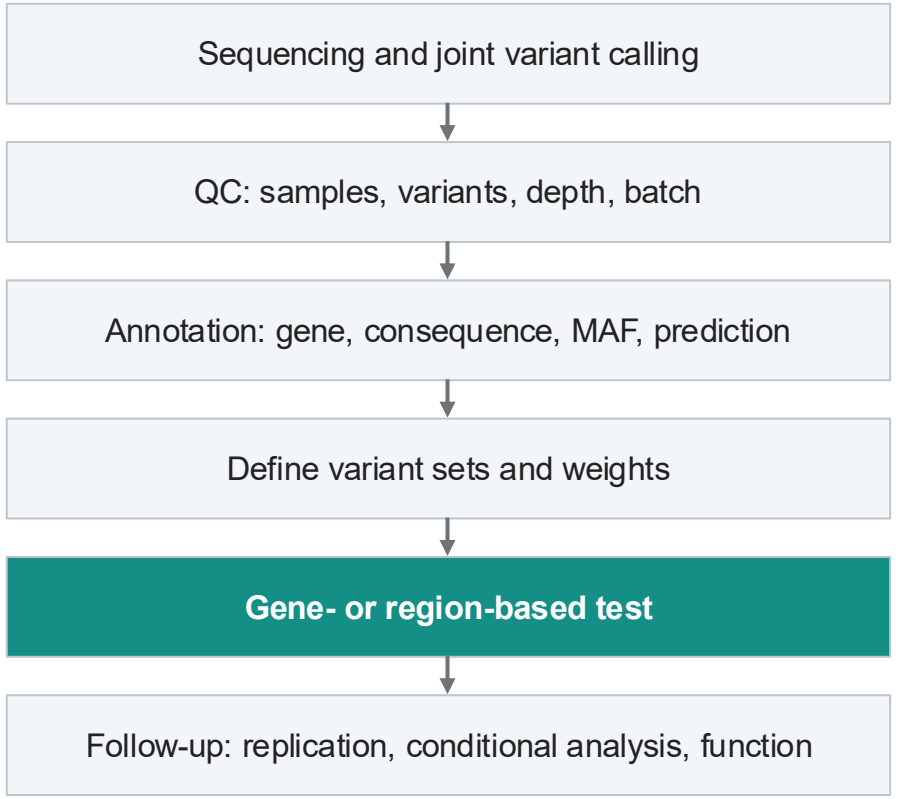
Single variant tests are not a feasible for majority of the GWAS studies because of the platforms, design, sample size etc.

Platforms: whole genome, exome, exome array, imputation; each trades coverage of rare alleles against cost

Designs considering the budget:

- extreme-phenotype sampling,
- two-stage sequencing then genotyping,
- family-based sampling to enrich rare alleles

Analytical considerations: joint calling, QC, ancestry and relatedness should be considered beforehand



Analysis flow for a sequencing association study (after Box 1)

Power comes from sample size and design; the test can only make the most of what was collected

Lee, Abecasis, Boehnke & Lin (2014) Am J Hum Genet

Families of rare variant tests

Family	Idea	Examples	Most powerful when
Burden	Collapse variants into one score	CAST, CMC, WSS (Madsen–Browning), MZ	Most variants causal, same direction
Adaptive burden	Let the data pick thresholds, weights or directions	aSum, VT, KBAC, EREC, step-up	Same as burden, but robust to some neutral or opposite alleles; often need permutation
Variance component	Test the variance of per-variant effects	SKAT, C-alpha, SSU	Mixed directions, or few causal among many
Combined (omnibus)	Mix burden and variance-component statistics	SKAT-O, Fisher / minimum-p combinations, MiST	Architecture unknown
Exponential combination	Exponentially weighted sum of scores	EC test	Very few causal variants with large effects; computationally heavy

All are **regression-based score tests**, so covariates, binary and continuous traits, and related samples are handled the same way

Additional considerations for a population based rare variant study

Getting the test right

Which variants: MAF cut-off and exact masks used change the answer. Try more than one mask.

Weights: rarer-is-stronger is used as a prior in most tests (true on average but not universally).

Small samples / low MAC. Asymptotic p-values are too liberal. Use moment-based adjustment or resampling

Case–control imbalance inflates type I error. Saddle-point or Firth corrections (SAIGE-GENE+)

Getting the study right

Population structure: PCs built from common variants do not fully correct rare-variant tests. Use mixed models and within-ancestry analysis

Relatedness: common in sequencing studies, especially families. Most methods account for kinship

Meta-analysis: share per-variant scores and their covariance, not just p-values, so burden, SKAT and SKAT-O can be rebuilt across studies (MetaSKAT, RAREMETAL, seqMeta)

Replication: can be difficult. The same gene (not the same variant) can be the unit of replication

Most false positives in rare-variant studies come from the data and the set definition, not from the test statistic

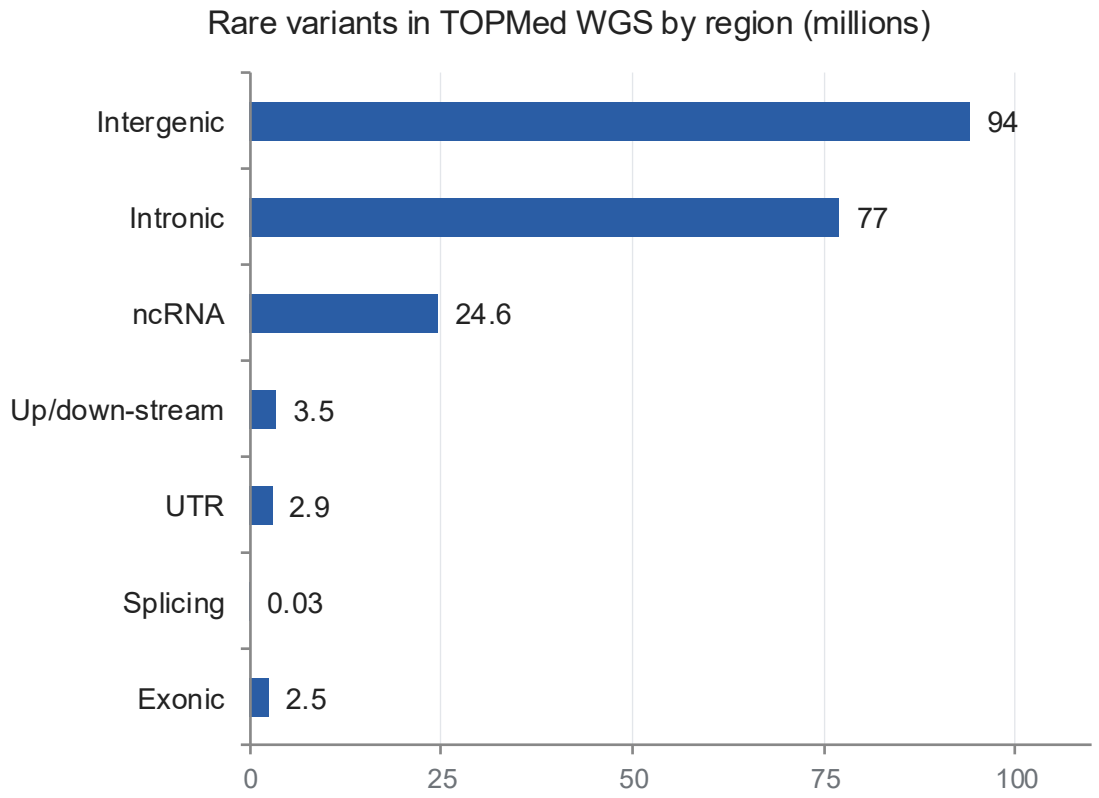
Rare variants outside exons

98.5% of the genome does not code for protein, and ~90% of GWAS signals sit there

WGS now gives us the rare variants in promoters, enhancers and non-coding RNAs too

Two new problems: there is **no natural unit** like a gene, and **most variants are neutral**

So, we need ways to define sets (elements, windows) and to tell likely functional variants from the rest



Coding variants are about 1% of what a genome sequence gives you

Non-coding rare variant findings

Element	Study	Finding
RNU4-2 (snRNA gene)	Burden over 41,132 non-coding genes; 5,529 ID cases vs 46,401 controls (100,000 Genomes)	De novo variants in an 18-bp region cause one of the most common monogenic NDDs (~0.4% of cases); invisible to exomes
Intergenic region near APOC1P1	STAAR, TOPMed: 12,316 discovery + 17,822 replication genomes	Rare variants in a 2-kb window associated with LDL-C, replicated
Long non-coding RNAs	STAAR, TOPMed: 66,329 genomes, 165,375 lncRNAs	83 lncRNA variant sets associated with lipids, all near known lipid genes; some independent of common variants
Regulatory variants near genes	LRT-q, GTEx: 681 genomes, 49 tissues	Nearby rare variants change expression of many genes; enriched in expression outliers

Signals are real but sparse. A few functional regulatory variants.

Rare variants and molecular phenotype: Proteome

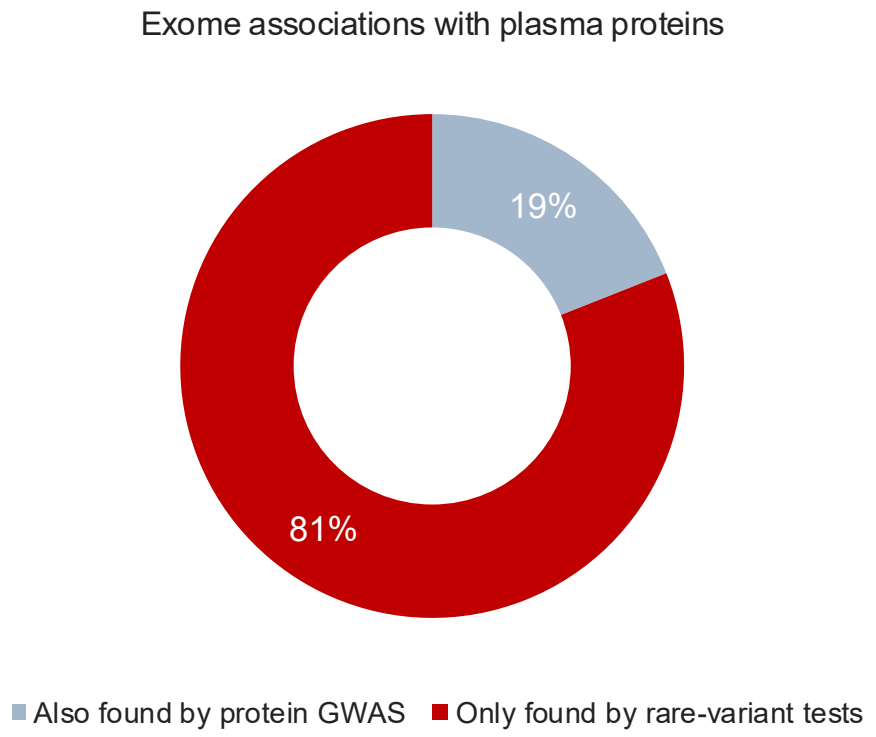
UK Biobank: exomes of **49,736** people with **2,923** plasma proteins (Olink)

Gene-level burden tests found **1,962** gene–protein associations

Many are **cis**: a gene's loss-of-function alleles lower its own protein, a built-in check that the method works

Trans associations link a gene to proteins elsewhere, pointing to pathways and possible drug effects

Most rare-variant hits were **not found** by protein GWAS of common variants



Proteins are an intermediate molecular phenotype: they tell us what a rare allele actually does

Non-coding alleles and using annotations: STAAR

Burden and SKAT weight variants by **frequency only**; a neutral and a damaging allele at the same MAF count the same

We have many annotations that predict function: conservation, CADD, open chromatin, histone marks, TF binding

They are correlated, so STAAR summarises each group into an **annotation PC** (aPC)

Each aPC becomes a **weight**: variants with stronger functional evidence count more

STAAR does not pick one annotation; it tries several and lets the data decide which helps

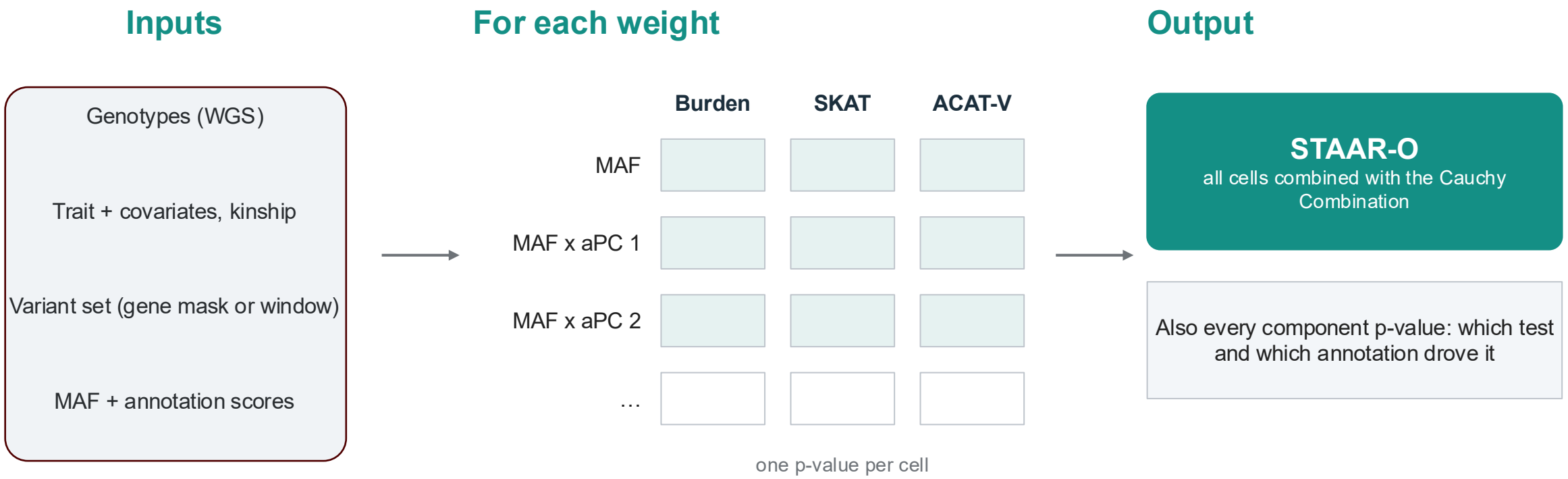
Annotation group	Examples
Conservation	GERP, phyloP, LINSIGHT
Protein function	CADD, PolyPhen, SIFT
Epigenetics	DNase, H3K27ac, H3K4me1/3
Transcription factors	TF binding (ENCODE)
Local context	mappability, distance to TSS



one aPC per group = one functional weight per variant

Li X et al. (2020) Nat Genet

STAAR: what goes in, what comes out



Intuition: run the familiar tests many times, each time **trusting a different annotation**, then combine

Burden covers dense same-direction effects, SKAT mixed directions, ACAT-V a few strong variants

The Cauchy combination is valid under correlation and needs no permutation, so it scales to biobanks

Reading STAAR output

NPC1L1, disruptive missense	Burden	SKAT	ACAT-V
MAF weight	2e-4	3e-3	1e-2
MAF x protein-function aPC	**4e-8**	6e-5	2e-4
MAF x epigenetic aPC	5e-3	1e-2	4e-2
STAAR-O		**1e-7**	

STAAR-O answers one question: is this set associated?

Which test is smallest hints at architecture: burden = same direction, ACAT-V = a few strong variants

Which weight is smallest says which functional class carries the signal

Then **condition** on nearby GWAS variants to show the signal is not a shadow of a common one

It does **not** say which variant is causal

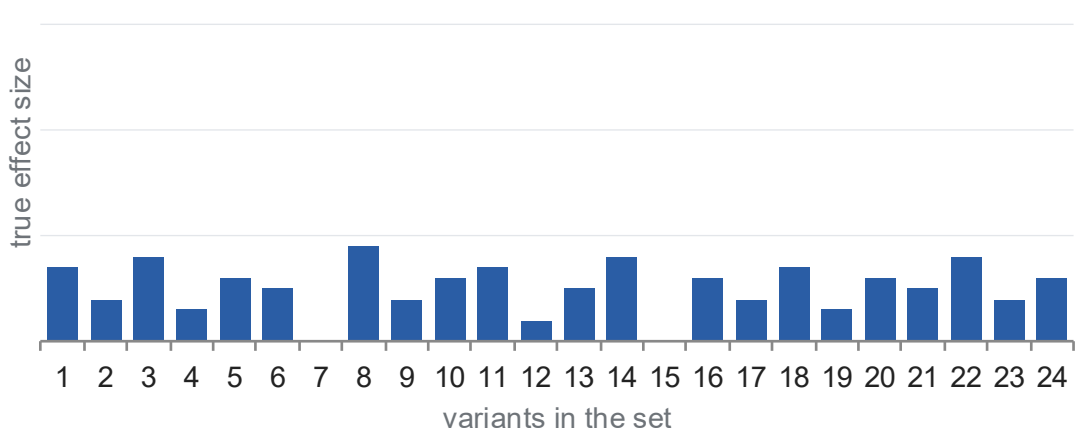
The p-value says 'this set'; the components say 'why'

Resources for annotating and interpreting variants

Resource	What it gives you	Where it enters the analysis
gnomAD	Population allele frequencies; gene constraint (pLI, LOEUF)	MAF cut-offs for masks; **which genes are intolerant** of damage
VEP, ANNOVAR, SnpEff	Consequence of each variant: synonymous, missense, stop-gain, splice, UTR	Defines the **variant classes** used to build masks
LOFTEE	Flags low-confidence predicted loss-of-function calls (e.g. last-exon truncations)	Keeps a LoF mask **truly LoF**
CADD, REVEL, AlphaMissense	Deleteriousness scores, especially for missense variants	Selects **damaging missense**; scores can also act as weights
FAVOR	Hundreds of annotations for every possible SNV and observed indel, plus precomputed annotation PCs	Source of STAAR **aPC weights** and noncoding categories
ENCODE, Roadmap, GeneHancer	Regulatory elements: promoters, enhancers, open chromatin, histone marks	Defines **noncoding variant sets** for whole-genome analysis

Dense versus sparse signals

Dense: most variants have small effects



Sparse: one or two variants with large effects



Sum-based tests add evidence over every variant

Burden, SKAT, SKAT-O

Ideal when the signal is spread out; in a sparse set the many nulls add noise and drown it

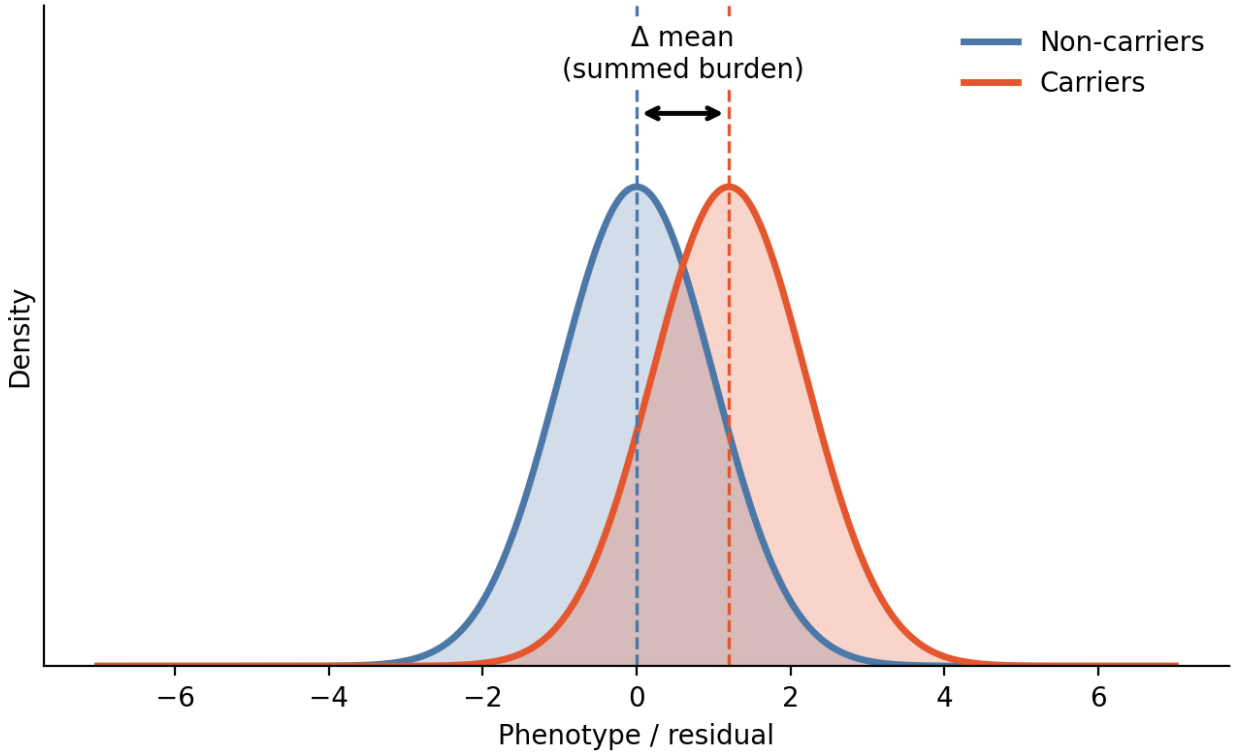
Tail-based tests look only at the strongest few

ACAT-V, Higher Criticism / GHC, GBJ, LRT-q, minP

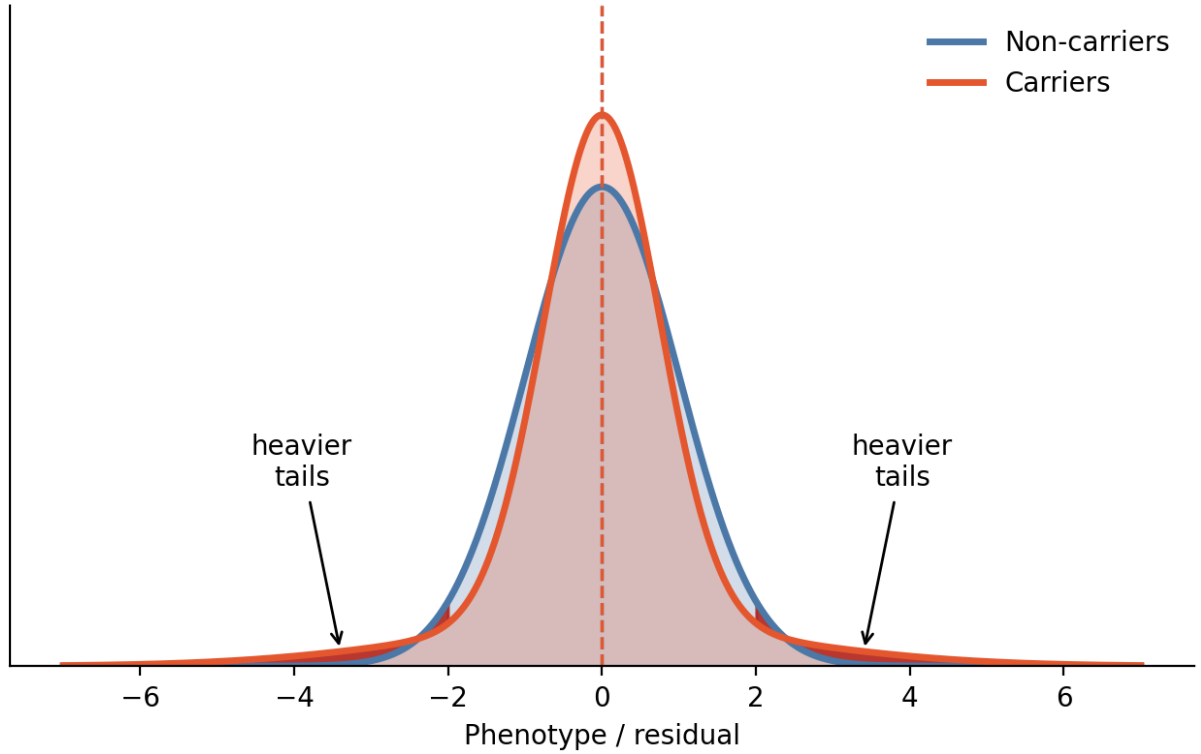
Ideal for large non-coding sets or windows where only a handful of variants matter

Non-coding sets are usually sparse, which is why sparse tests matter for WGS

Two different scenarios



Dense signal: Mean-based tests



Sparse signal: Tail-based tests

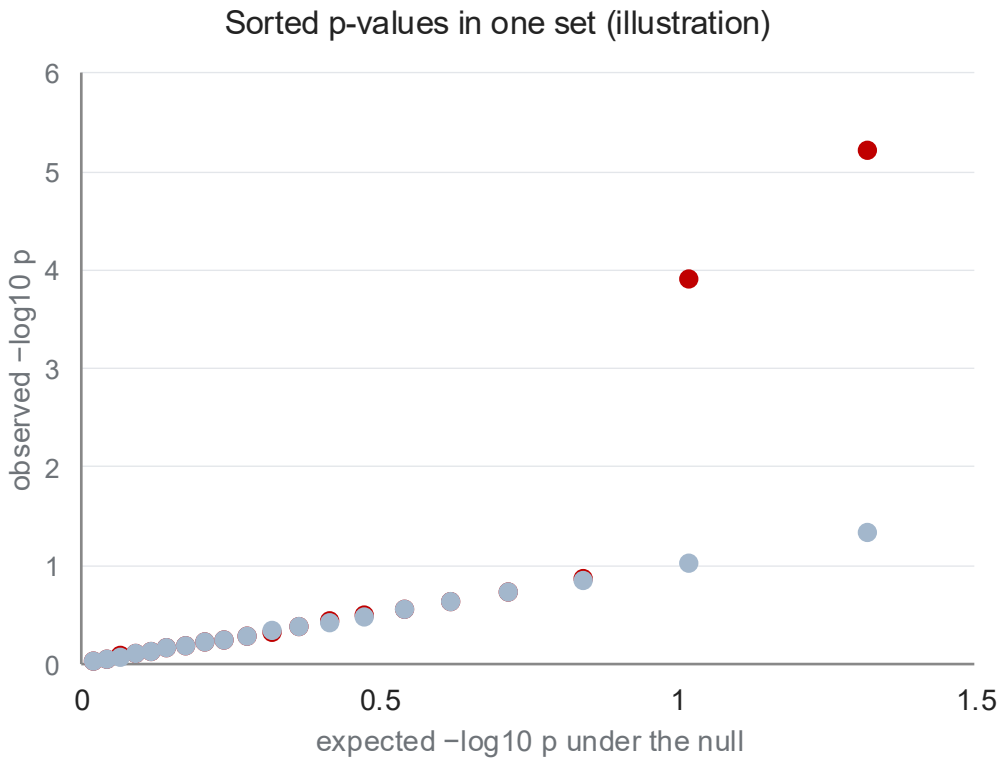
How tail-based tests work

Sort the per-variant p-values and ask: are there **more small p-values than chance** would give?

Higher Criticism: at each threshold compare the observed count of p-values below it with the expected count; take the largest standardised excess

GHC adds LD and small set sizes; **GBJ** uses a likelihood ratio, better when signals are moderately sparse

LRT-q infers which variants are likely causal and builds a likelihood ratio from only those



18 variants follow the null; 2 stand out in the tail

Several traits at once: why and where it has worked

Rare variants in one gene often move **several related traits** (pleiotropy)

Correlated traits carry shared information. A joint test can **gain power** over testing each one

It also asks a question single-trait tests cannot: does this gene affect a **pattern or group** of traits?

Multi-SKAT on METSIM amino acids and exome-array data: more power than trait-by-trait SKAT-O

MultiSTAAR on LDL, HDL, TG in 61,838 TOPMed genomes: new coding and non-coding associations missed by single-trait analysis

LoF carriers in	LDL	TG	HDL
PCSK9	↓	–	–
APOC3	–	↓	↑
ANGPTL3	↓	↓	↓
LPL	–	↑	↓

Multi-trait rare variant methods

Method	Idea	Data
GAMuT	Compares similarity of people in genotypes with their similarity across all traits (kernel distance-covariance)	Individual-level
Multi-SKAT	SKAT with a second kernel for how a variant's effects relate across traits; omnibus over kernels; handles kinship	Individual-level
Meta-MultiSKAT	The same idea from per-study score summaries	Summary statistics
MTAR	Multi-trait burden, SKAT and omnibus tests; allows overlapping samples between traits	Summary statistics
MultiSTAAR	Multi-trait version of STAAR: annotation weights, coding and non-coding sets, WGS scale	Individual-level (WGS)

The key assumption: effects across traits

Homogeneous
same effect on every trait

Heterogeneous
effects unrelated across traits

Phenotype-correlated
effects follow the trait correlation

Orthogonal
effects on independent trait axes

Data, browsers and software for rare-variant analysis

Type	Resource	What it is for
Data	**UK Biobank, All of Us, TOPMed**	Large exome and whole-genome sequencing cohorts
Browser	**Genebass**	Gene-based results across UK Biobank exomes: look up a gene or trait
Browser	**STAAR PheWAS portal**	Rare coding and noncoding results from UK Biobank genomes
Software	**SKAT / SKAT-O, rvtests**	Classic burden, SKAT and SKAT-O tests; good for learning
Software	**SAIGE-GENE+, REGENIE**	Biobank-scale gene tests; handle relatedness and unbalanced case-control data
Software	**STAAR, STAARpipeline**	Annotation-weighted tests for exomes and genomes, coding and noncoding
Meta-analysis	**MetaSTAAR, Meta-SAIGE**	Combine rare-variant results across cohorts from summary statistics
Multi-trait	**Multi-SKAT, MultiSTAAR**	Test one gene against several correlated traits jointly

Thank you



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